

EDS THEORY ASSIGNMENT: 2

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Dataset: WordNet Lexical Database

1. Checking Missing Values

Problem: Identify null values in the dataset.

Solution:

```
df.isnull().sum()
```

2. Counting Total Words

Problem: Find the total number of words in the dataset.

Solution:

```
total_words = df['word'].count()
```

3. Finding Unique Words

Problem: Count unique words present in WordNet.

Solution:

```
unique_words = df['word'].nunique()
```

4. Synset Distribution

Problem: Count the number of synsets for each part of speech.

Solution:

```
df['pos'].value_counts()
```

5. Average Synsets per Word

Problem: Calculate average synsets associated with a word.

Solution:

```
df['synset_count'].mean()
```

6. Maximum Synset Count

Problem: Find the word with the highest number of synsets.

Solution:

```
df.loc[df['synset_count'].idxmax()]
```

7. Minimum Synset Count

Problem: Find the word with the least number of synsets.

Solution:

```
df.loc[df['synset_count'].idxmin()]
```

8. Standard Deviation of Synsets

Problem: Measure variation in synset counts.

Solution:

```
np.std(df['synset_count'])
```

9. Filtering Nouns

Problem: Extract all noun entries from WordNet.

Solution:

```
nouns = df[df['pos'] == 'noun']
```

10. Filtering Verbs

Problem: Extract all verb entries from WordNet.

Solution:

```
verbs = df[df['pos'] == 'verb']
```

11. Longest Definition

Problem: Find the word having the longest definition.

Solution:

```
df.loc[df['definition'].str.len().idxmax()]
```

12. Average Definition Length

Problem: Calculate average definition length.

Solution:

```
df['definition'].str.len().mean()
```

13. Word Length Analysis

Problem: Add a column showing length of each word.

Solution:

```
df['word_length'] = df['word'].apply(len)
```

14. Most Frequent Word Length

Problem: Find the most common word length.

Solution:

```
df['word_length'].mode()
```

15. Correlation Analysis

Problem: Determine correlation between word length and synset count.

Solution:

```
df['word_length'].corr(df['synset_count'])
```

16. Sorting by Synset Count

Problem: Arrange words in descending order of synsets.

Solution:

```
df.sort_values('synset_count', ascending=False)
```

17. Duplicate Detection

Problem: Identify duplicate words in the dataset.

Solution:

```
df[df.duplicated('word')]
```

18. Top 5 Words by Synset Count

Problem: Display five words having the highest synset counts.

Solution:

```
df.nlargest(5, 'synset_count')
```

19. Grouping by Part of Speech

Problem: Calculate average synset count for each part of speech.

Solution:

```
df.groupby('pos')['synset_count'].mean()
```

20. Random Sampling

Problem: Select five random records for inspection.

Solution:

```
df.sample(5)
```